(Please write your Enrollment Number) Enrollment No._____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech. (CSE/IT), Semester:6th
(April -May, 2023)

Subject Code: BCS 306	Subject: Compiler Design

Time : 3 Hours Maximum Marks :60

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 (2.5*8=20)

(a) What is the first phase in Compiler? How a token is recognized, explain with an example?

(b) What is left factoring in a Grammar? Remove left factoring from following grammar:
S→iEtS/ iEtSeS/a
E→b

(c) Eliminate left recursion in the following grammar:
A → ABd | Aa | a
B → Be | b

(d) Define Context Free Grammar (CFG)? Write about Derivation trees and Parse tree.

(e) Define FIRST and FOLLOW functions, explain with an example Grammar.

(f) Discuss about bootstrapping with a neat diagram?

(g) Define Ambiguity? Check whether the grammar S→aAB, A→bC/cd, C→cd, B→c/d, is Ambiguous or not?

(h) What is the second phase in Compiler? How grammar is useful in second phase?

UNIT-I

Q2 Explain in detail about various phases of a compiler with a neat diagram and suitable example for each phase? (10)

Q3 Discuss in detail the uses of Finite automata, Regular expression and its applications in lexical analysis with an example? (10)

UNIT-II

Q4 Construct LR(0) items and SLR Parsing table for the grammar S→L=R/R, L→*R/id, R→L (10)

Q5 Construct the LALR parse table for the following grammar. (10)
S→aB/aC/Sd/Se

B→bBc/f

C→g

UNIT-III

Q6 Define what Syntax directed definitions is, and explain about following with example. (10)
(i). Bottom-up evaluation of S- attributed Definitions
(ii). Bottom-up evaluation of inherited attributes.

Q7 Write about different Source Language issues, and explain about Storage allocation strategies? (10)

UNIT-IV

Q8 Describe and explain about the following with suitable example: (10)
(i). Abstract syntax tree and polish notation
(ii). Three Address code
(iii). Quadruples
(iv). Triples.

Q9 What is code optimization? Explain in details with example about Principal sources of optimization. (10)